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Arnaud Blouin & Benoit Combemale (Univ Rennes, IRISA, Inria)
Romain Rouvoy (University of Lille)

Subject: Application for PhD Position – Modeling the construction of digital twins (PC3)

Dear Arnaud Blouin, Benoit Combemale, and Romain Rouvoy,

I am applying with great enthusiasm for the PC3 PhD position focused on "Modeling the construction of digital twins." Given my deep engineering background in systems architecture, model-driven engineering (MDE), and continuous deployment, I am highly motivated to define the formal methodologies required to orchestrate the lifecycle of complex, multi-layered digital twin systems.

The industrial deployment of a Digital Twin is rarely a single, monolithic event; it is a continuous process of integration, validation, and updating. During my work engineering the "Autonomous Cyber-Shield Platform," I confronted the complexities of continuous integration for cyber-physical models. I architected a deployment pipeline that synchronized updates between a Python-based Reinforcement Learning agent and a live Next.js visualization dashboard, all without disrupting the core system runtime. This required defining strict configuration models and deployment semantics. I am eager to formally abstract these practices into generalized modeling languages. My goal is to research and formalize the construction processes—defining exactly how simulation models, data pipelines, and UI components are assembled and deployed dynamically across the cloud-to-edge continuum.

Furthermore, my expertise in building highly modular architectures positions me perfectly to contribute to the Artemis framework. I am excited to collaborate with the DiverSE and Spirals teams to build the robust software engineering tooling needed to turn digital twin construction from an ad-hoc craft into an automated, mathematically sound discipline. Please find attached my resume and academic transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui